



MG-1P Functional Module



Overview



MG-1P Main Controller



The MG-1P main controller mainly coordinates the normal operation of each functional module of the aircraft

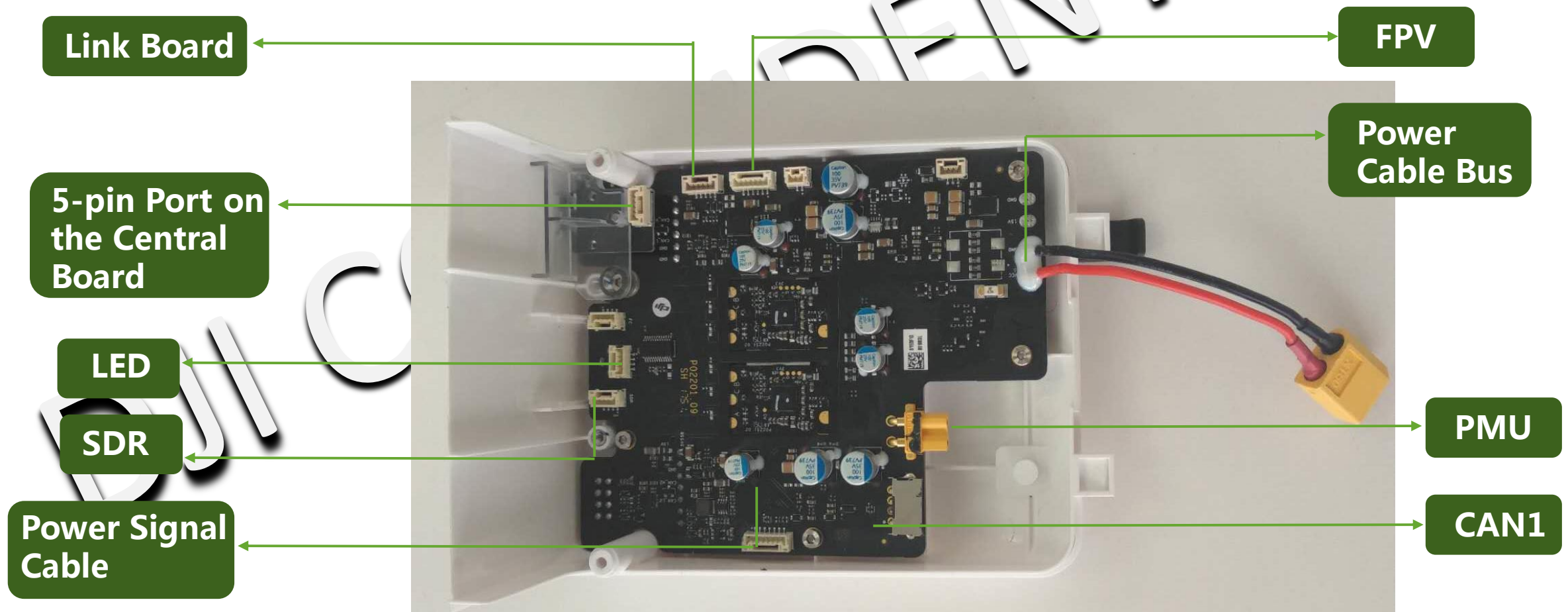


- **M1-M8:** Connects M1-M8 ESC cable.
- **M1, M3, M5, and M7** ESCs are white and grey, while **M2, M4, M6, and M8** ESC cables are white and black. The M1 and M8 ESC cables are on the same arm.
- **CAN1:** Connects the central board.
- **CAN2:** Connects the GPS-RTK CAN1 port.
- **iESC:** Connects the signal cable and four ESC feeders to the 485 adaptive board.
- **RF:** Connects the SDR module to the SDR port on the central board.
- **LED:** Connects the 4-pin USB port on the central board.

Center Board

The center board is mainly a power conversion module, which converts the 12S power to the voltage required for each module (PMU is on the center board, and the PMU function is unchanged).

The link board is connected to the 5-pin port on the center board, which is the link button module on the SDR module.



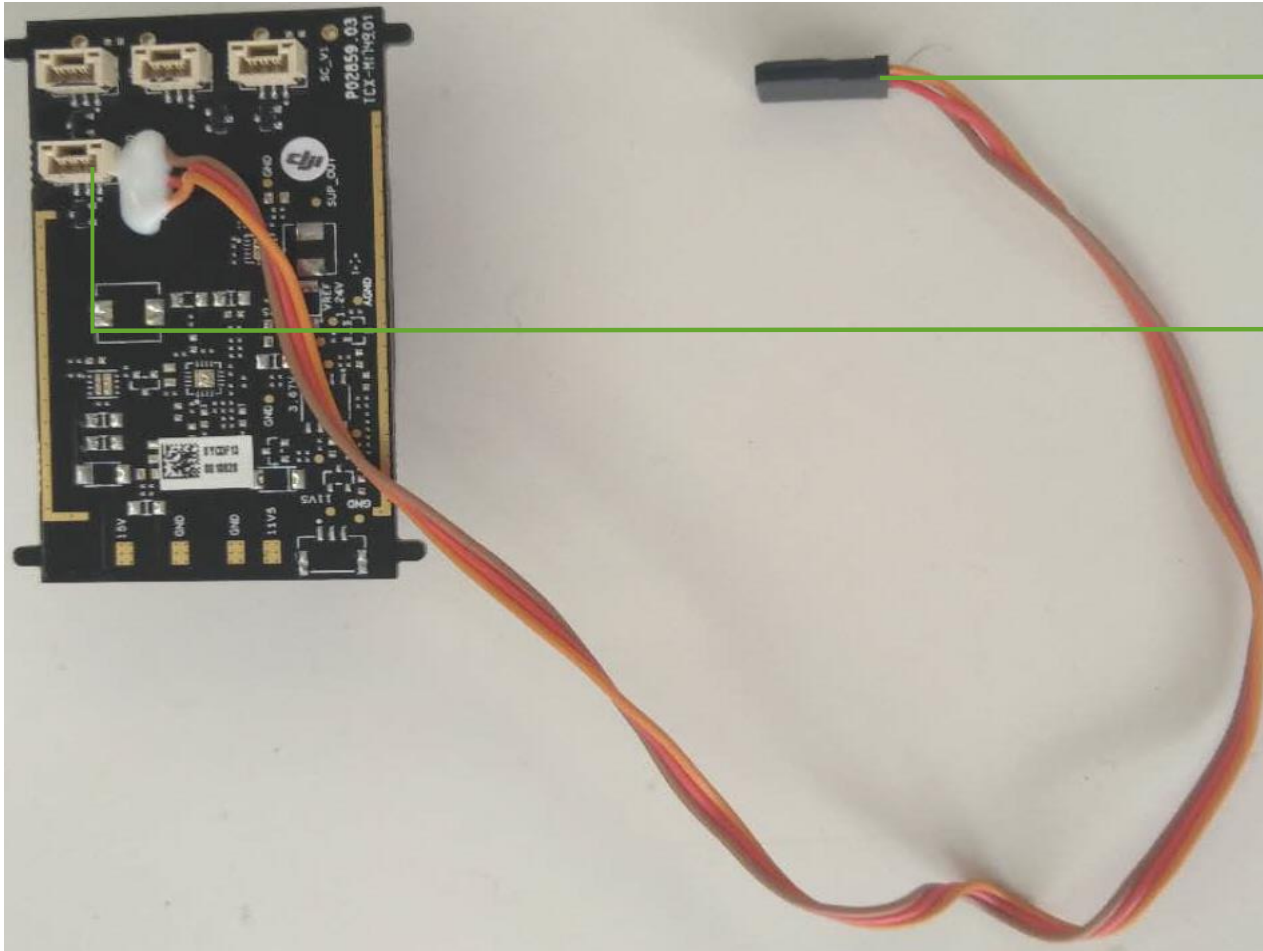


GPS: Global Positioning System
MG-1P can locate and identify
the direction with the GPS
module and compass module.

Compass

CAN2

485 Adaptive Board



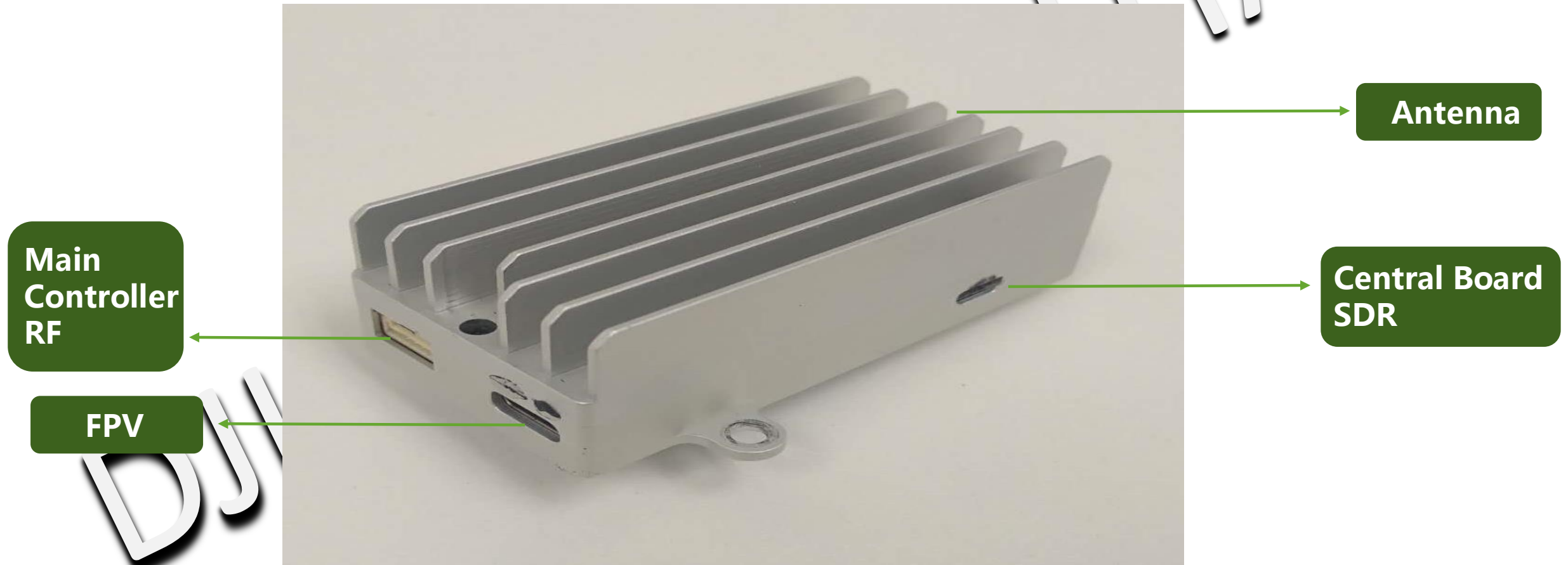
IESC

ESC Feeder

- The 485 adapter board receives the signal of the four ESC feeders and reports them to the main controller.
- The J1, J2, J3, and J4 ports on the 485 adapter board are the same.

SDR Module

The SDR module communicates with the remote controller and transmits the FPV image transmission.

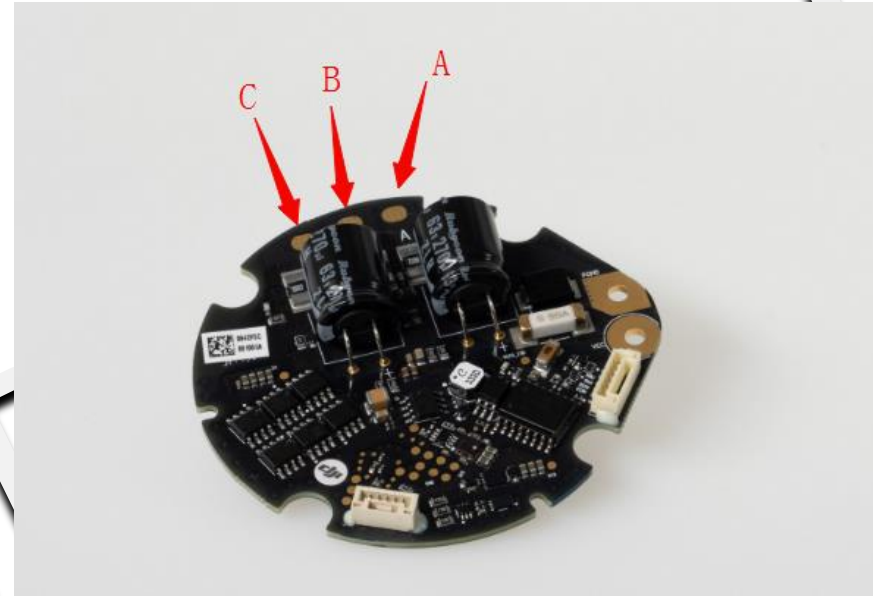


FPV + Camera

- The FPV camera allows you to see where the drone is flying and what it is seeing as if it had its own eyes, which helps the drone avoid obstacles when the drone flies far away. It can also send the live video in real-time, helping the drone record waypoints in the air.
- Searchlight can better observe the surrounding environment and ensure efficient and safe operation.



Motors and ESCs

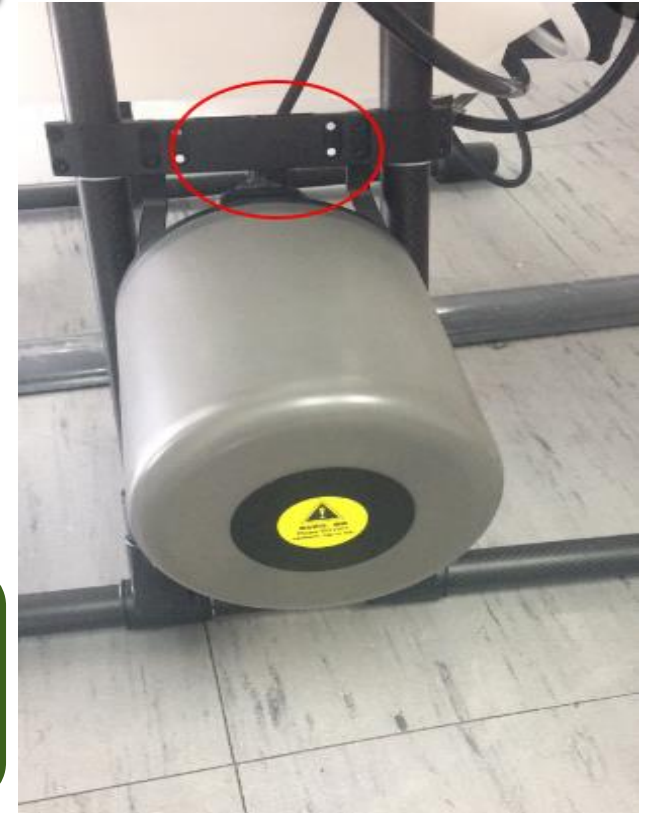


1. Connection between M1, M3, M5, and M7 motors and ESCs: The red motor cable needs to connect to A, black to B, and blue to C.
2. Connection between M2, M4, M6, and M8 motors and ESCs: The red motor cable needs to connect to A, blue to B, and black to C. **The eight motors or ESCs are the same.**

High-Precision Radar

- The MG-1P's improved radar system integrates the three high-precision terrain detection radars with an obstacle avoidance radar. This doubles detection sensitivity and allows for multi-directional detection, significantly increasing the aircraft's ability to sense obstacles and follow terrain automatically;
- The drone can detect the distance between the aircraft and the front slope, the bottom crop, and the rear slope, check whether there are obstacles in front or to the rear of the drone, and report it to the main controller in real-time.

**Correct Installation
Diagram**



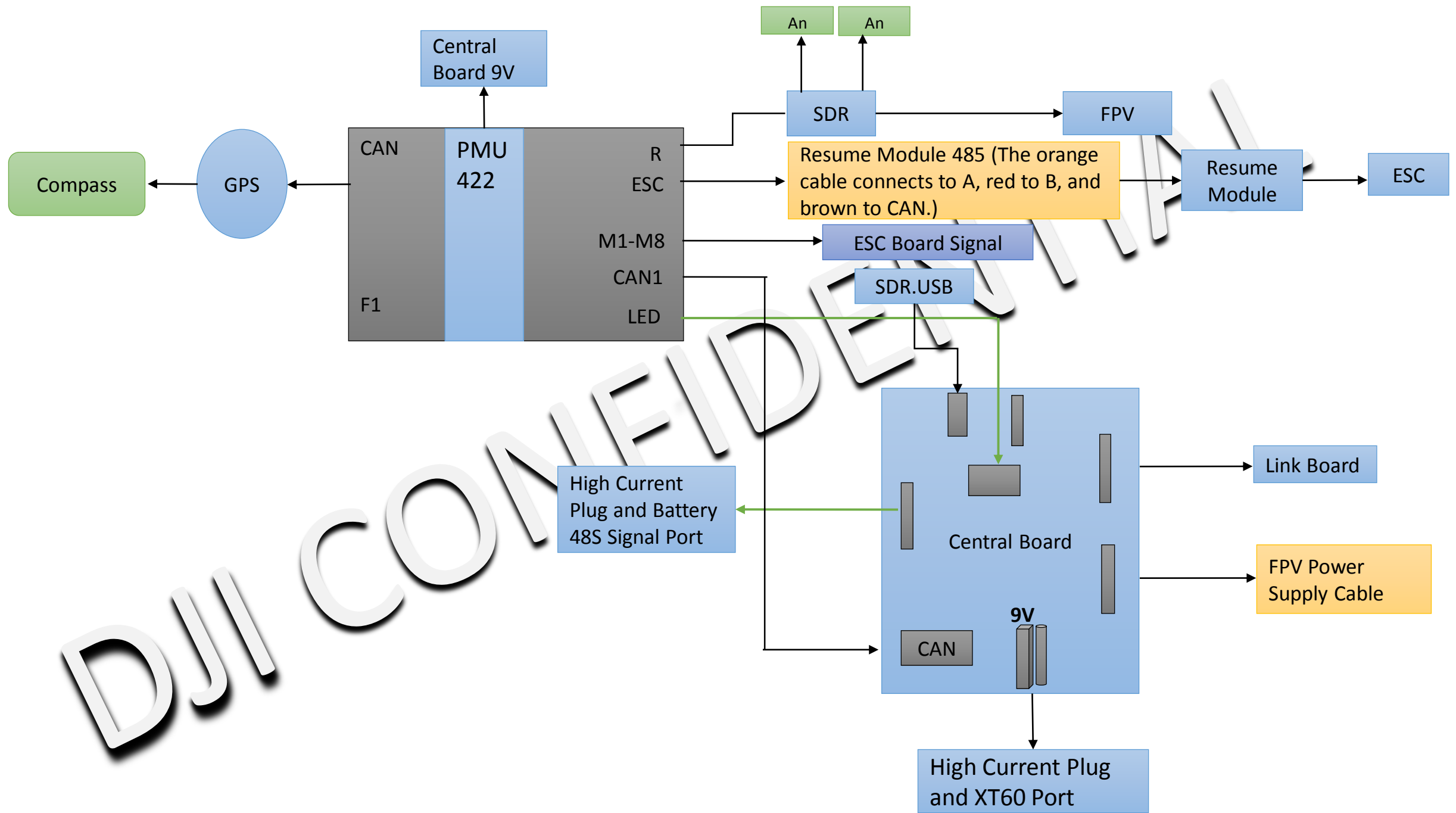
**Connecting the
Central Board**

Delivery Pump

The water outlet on the delivery pump has been redesigned, and the left and right pumps are the same. The diaphragm material and process have been upgraded to improve the corrosion resistance. The connection between the delivery pump and the liquid tank and water pipeline is optimized, and it is easier to be disassembled.



**Connecting the
Central Board**





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**THANK
YOU**

DJI